DHAIRYA BHATIA

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SUMMARY

ML researcher studying whether video-language models understand motion. Builds video benchmarks, evaluation pipelines, and multimodal datasets pairing video with wearable sensor signals. Currently extending MotionBlind with a larger dataset.

EDUCATION

Khoury College of Computer Sciences, Northeastern University — Boston, MA
Master of Science in Computer Science — *Sep 2025 – May 2027 (Expected)*

The University of British Columbia — Kelowna, BC, Canada
Bachelor of Science in Computer Science — *Sep 2020 – May 2024*

RESEARCH EXPERIENCE

Augmented Cognition Lab (ACLab), Northeastern University — May 2026 – Present
Research Assistant · Advisor: Prof. Sarah Ostadabbas

- Co-led MotionBlind, a paired-clip benchmark for motion understanding in Video-LLMs.
- Built clip pairs that differ only in motion, such as speed, magnitude, or direction.
- Found the best open model reaches 11.7% paired accuracy versus 91.3% for humans.
- Showed that scaling, more frames, and better frame selection do not close the gap.
- Designed a human study with five reviewers.
- Built SLURM evaluation pipelines for open-weight and API-based models.
- Collecting egocentric and exocentric video paired with wrist-worn SNC signals.
- Extending MotionBlind with a larger dataset.
- Contributed to the CV4Smalls CVPR 2026 challenge dataset.

CACTILab, Northeastern University — Mar 2026 – Present
Research Volunteer

- Built a multi-agent LLM pipeline that generates embedded firmware.
- Added compiler-in-the-loop verification of generated code.
- Implemented model tiering and per-turn cost tracking.
- Delivered a working MVP on a TI MSPM0L2228 board.

Directed Study, Northeastern University — Fall 2026
Advisor: Prof. Sudhakaran Prabakaran

- Applying machine learning to multi-omics expression data.

PUBLICATIONS & PREPRINTS

Dhairya Bhatia, Bishoy Galoaa, et al. *MotionBlind: Probing the Illusion of Motion Understanding in Video-LLMs.* arXiv preprint arXiv:2609.09528, 2026. Co-first author.

TEACHING EXPERIENCE

Teaching Assistant, CS3100, Northeastern University — Jan 2026 – Jun 2026

- Supported students in labs and office hours.
- Helped design and test labs and assignments before release.
- Led an analysis of student feedback on AI-assisted programming workflows.

INDUSTRY EXPERIENCE

Oriv — Sep 2024 – Jul 2025
Software Development Intern / AI Developer

- Prototyped knowledge-graph methods for IoT sensor signals in industrial automation.
- Introduced sensor virtualization to simulate data without physical hardware.

PROJECTS & COMPETITIONS

ECG Diagnosis on PTB-XL (Independent Project)

- Trained PyTorch classifiers on 12-lead ECG data using SLURM.
- Reached 0.922 macro AUROC over 3 seeds.
- Reached 0.835 macro AUROC on an external dataset.

MITRE eCTF 2026, Embedded Security Competition

- Contributed firmware implementation, code review, and bug fixes.

Project Escalion, NU Launch Labs (Co-Founder & Team Lead)

- Co-founded a Rust-based game studio, now a Khoury-recognized club.
- Designing a custom engine for high-performance rendering.

TECHNICAL SKILLS

Languages: Python, Rust, C, Java, SQL, Bash
ML / AI: PyTorch, Multimodal Learning, Video-Language Models, Model Evaluation & Benchmarking, Computer Vision
Signal & Data: Signal Processing, Time-Series and Sensor Data, Multimodal Data
Systems & Tools: Linux, SLURM, HPC, Docker, Git, Embedded Systems, Cloud Deployment